- ***3*** Maps the action outputs of the DQN to a subset of actions in the environment

The `update_replay` function adds new memories to the experience replay if it is not yet full; if it is full, it will replace random memories with new ones. If the memory resulted in a positive reward, we add 50 copies of that memory, since positive reward memories are rare and we want to enrich the experience replay with these more important memories.

All the MiniGrid environments have seven actions, but in the environment we will use in this chapter, we only need to use five of the seven actions, so we use a dictionary to translate from the output of DQN, which will produce actions 0–4, to the appropriate actions in the environment, which are {0,1,2,3,5}.

The MiniGrid's action names and corresponding action numbers are listed here:

```
[<Actions.left: 0>,
 <Actions.right: 1>,
 <Actions.forward: 2>,
 <Actions.pickup: 3>,
 <Actions.drop: 4>,
 <Actions.toggle: 5>,
 <Actions.done: 6>]
```

In listing 10.10 we jump into the main training loop of the algorithm.

**Listing 10.10. The main training loop**

```
from collections import deque
env = ImgObsWrapper(gym.make('MiniGrid-DoorKey-5x5-v0'))
1
state = prepare_state(env.reset())
GWagent = MultiHeadRelationalModule()
2
Tnet = MultiHeadRelationalModule()
3
maxsteps = 400
4
env.max_steps = maxsteps
env.env.max_steps = maxsteps
```